LAB 3 – ARCHITECTURE SELECTION & JUSTIFICATION

Freelance Content Creator Escrow Platform (Problem Statement #53)

Architecture Selection: We chose a **Layered Architecture**, with modular business components, for the Freelance Content Creator Escrow Platform.

Architectural style analysis. Layered: separates presentation, business rules and data/integration, making the milestone workflow easier to maintain and test; its main drawback is possible cross-layer overhead. **Microservices:** allows independent deployment and scaling of contract, watermark, review and payment services, but adds operational complexity, network latency and distributed consistency concerns. **Client-Server:** centralizes business logic and data for straightforward control, but can create a server bottleneck and single point of failure. For this platform's current academic-scale scope and closely coordinated escrow transactions, Layered Architecture offers the best balance of clarity, transaction coordination and implementation simplicity.

Two reasons. (1) The workflow has clear stages—contract definition, draft submission, sponsor review, conditional payment release and dispute handling—so separating user interaction, business rules and persistence gives each responsibility a clear place. (2) Milestone and escrow operations are tightly related; a layered modular design avoids the overhead of independently deploying and coordinating many services while preserving modularity.

Security advantage. Escrow operations are mediated by application-layer services rather than allowing UI modules to manipulate balances directly. The escrow service enforces sponsor approval before release, freezes funds when a dispute is raised after two rejections, and records lock/release/refund events in an append-only audit log. Escrow wallet data is encrypted, consistent with NFR-002.

Performance benefit. Watermark processing is isolated as a business capability, allowing image/video preview processing to be optimized without changing contract, review or payment logic. The processing path can be measured against NFR-001's five-second target. Clear layer boundaries also reduce unnecessary cross-module work during ordinary review and approval.

Trade-off and diagram scope. Layered designs can add calls across layers and may be less independently scalable than microservices. For the current scope, simpler deployment, transaction coordination and maintainability outweigh independent service deployment. The accompanying diagram is a component-level view within the selected layered architecture; its components are logical modules, not necessarily separately deployed microservices.